Incompressible Flow Solver

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1 Cartesian Mesh and Governing Equations

In this section, we will derive the discretized operators for a 2D problem flow configuration over a cartesian mesh (shown in figure 1). The equations will be spatially discretized using finite volumes on a fully staggered grid to ensure primary conservation and to avoid pressure checkerboarding.

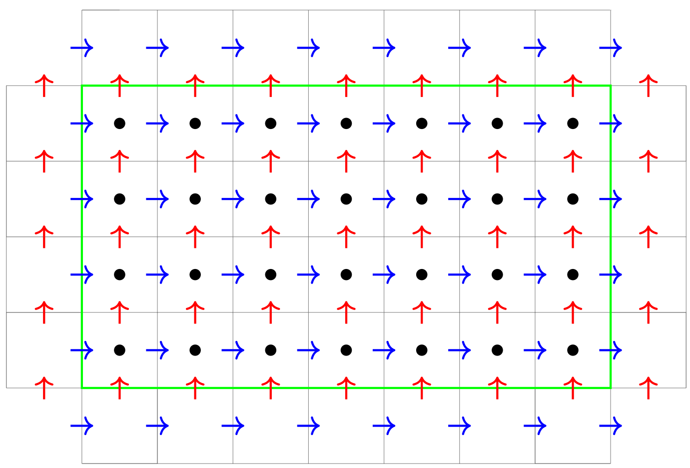

Figure 1: Staggered grid. The green line denotes the boundaries of the physical domain. Velocity and pressure values *inside* the blue domain are active degrees of freedom that must be solved for. Velocities *on* or *outside* the blue lines are boundary/ghost points that can be determined algebraically given the boundary conditions and the interior values. Blue arrows indicate x-velocities, red arrows y-velocities, and black dots denote pressure.

Before proceeding we recall the conservative form of the incompressible Navier-Stokes along with continuity.

$$\partial_t \mathbf{u} + \nabla \cdot \left(\mathbf{u}\mathbf{u}^\top\right) = -\nabla p + Re^{-1}\nabla^2 \mathbf{u} \tag{1}$$

$$\nabla \cdot \mathbf{u} = 0 \tag{2}$$

The integral form of the eqautions is obtained by integrating over the volume. The divergence theorem is used to turn the volume integrals into surface integrals, where $\hat{\mathbf{n}}$ denotes the outward-pointing, unit-norm vector normal to the surface. (In our 2D setting over a cartesian grid, we have $\hat{\mathbf{n}} = (1, 0)$ or $\hat{\mathbf{n}} = (0, 1)$.)

$$\int_{\mathcal{V}} \partial_t \mathbf{u} \, \mathrm{d}\mathcal{V} + \int_{\mathcal{S}} \mathbf{u} \left(\mathbf{u} \cdot \hat{\mathbf{n}} \right) \, \mathrm{d}\mathcal{S} = - \int_{\mathcal{S}} p \hat{\mathbf{n}} \, \mathrm{d}\mathcal{S} + Re^{-1} \int_{\mathcal{S}} \hat{\mathbf{n}} \cdot \nabla \mathbf{u} \, \mathrm{d}\mathcal{S} \tag{3}$$

$$\int_{\mathcal{S}} \mathbf{u} \cdot \hat{\mathbf{n}} \, \mathrm{d}\mathcal{S} = 0. \tag{4}$$

1.1 x-momentum equation in conservative form

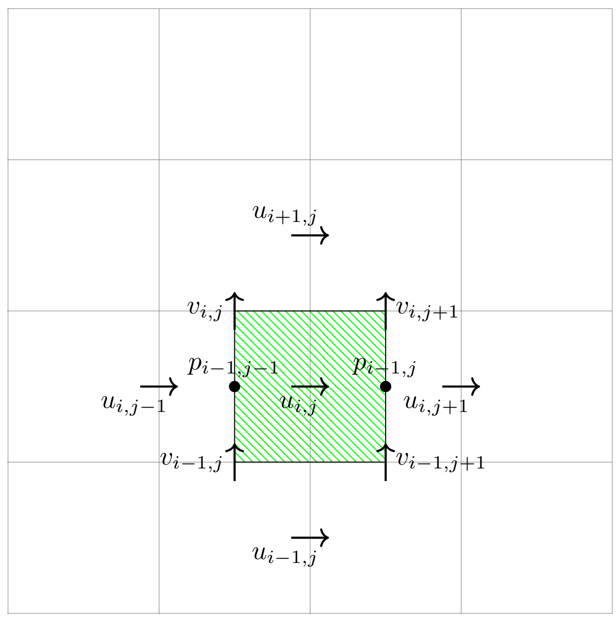

Figure 2: Control volume for x-momentum. The grid spacing in the x direction is Δx, and Δy in the y direction.

Starting from equation (3) in the x direction, we have

$$\int_{\mathcal{V}} \partial_t u \, \mathrm{d}\mathcal{V} + \int_{\mathcal{Y}} u^2 \, \mathrm{d}\mathcal{Y} + \int_{\mathcal{X}} uv \, \mathrm{d}\mathcal{X} = - \int_{\mathcal{Y}} p \, \mathrm{d}\mathcal{Y} + Re^{-1} \left(\int_{\mathcal{Y}} \partial_x u \, \mathrm{d}\mathcal{Y} + \int_{\mathcal{X}} \partial_y u \, \mathrm{d}\mathcal{X} \right). \tag{5}$$

In discrete form, using the control volume in figure 2, we have

$$\begin{aligned}
\int_{\mathcal{V}} \partial_t u \,\mathrm{d}\mathcal{V} &= \partial_t u_{i,j} \Delta x \Delta y \\
\int_{\mathcal{Y}} u^2 \,\mathrm{d}\mathcal{Y} &= \frac{\Delta y}{4} \left(\left(u_{i,j+1} + u_{i,j}\right)^2 - \left(u_{i,j+1} + u_{i,j}\right)^2 \right) \\
\int_{\mathcal{X}} uv \,\mathrm{d}\mathcal{X} &= \frac{\Delta x}{4} \left(\left(u_{i+1,j} + u_{i,j}\right)\left(v_{i,j} + v_{i,j+1}\right) - \left(u_{i-1,j} + u_{i,j}\right)\left(v_{i-1,j} + v_{i-1,j+1}\right) \right) \\
\int_{\mathcal{Y}} p \,\mathrm{d}\mathcal{Y} &= \left(p_{i-1,j} - p_{i-1,j-1}\right) \Delta y \\
Re^{-1} \int_{\mathcal{Y}} \partial_x u \,\mathrm{d}\mathcal{Y} &= \frac{Re^{-1}}{\Delta x} \left(u_{i,j+1} - 2u_{i,j} + u_{i,j-1}\right) \Delta y \\
Re^{-1} \int_{\mathcal{X}} \partial_y u \,\mathrm{d}\mathcal{X} &= \frac{Re^{-1}}{\Delta y} \left(u_{i+1,j} - 2u_{i,j} + u_{i-1,j}\right) \Delta x
\end{aligned} \tag{6}$$

Putting this all together and dividing through by $\Delta x \Delta y$, we obtain

$$\boxed{\begin{aligned}
&\partial_t u_{i,j} + \frac{1}{4\Delta x} \left(\left(u_{i,j+1} + u_{i,j}\right)^2 - \left(u_{i,j+1} + u_{i,j}\right)^2 \right) \\
&+ \frac{1}{4\Delta y} \left(\left(u_{i+1,j} + u_{i,j}\right)\left(v_{i,j} + v_{i,j+1}\right) - \left(u_{i-1,j} + u_{i,j}\right)\left(v_{i-1,j} + v_{i-1,j+1}\right) \right) \\
&= -\frac{p_{i-1,j} - p_{i-1,j-1}}{\Delta x} + Re^{-1} \left(\frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta x^2} + \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta y^2} \right).
\end{aligned}}$$

1.2 y-momentum equation in conservative form

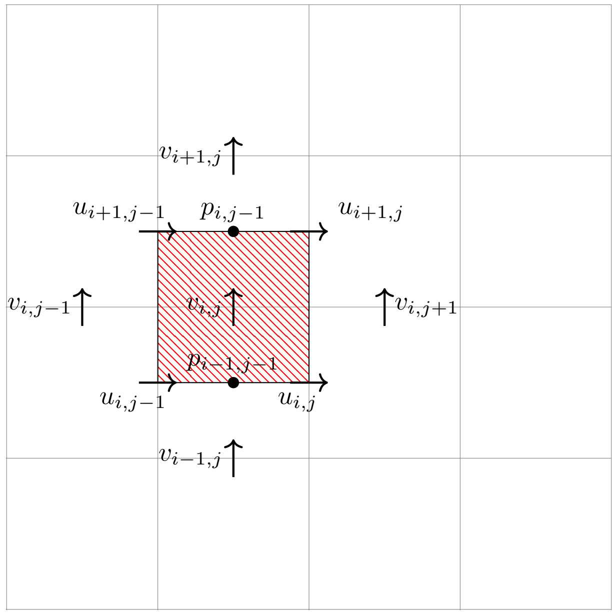

Figure 3: Control volume for y-momentum

We may recognize that the y-momentum is functionally symmetric to the x-momentum. It therefore follows that Equation (3) allows us to obtain

$$\underbrace{\int_{\mathcal{V}} \partial_t v \, \mathrm{d}\mathcal{V}}_{\text{A}} + \underbrace{\int_{\mathcal{Y}} v^2 \, \mathrm{d}\mathcal{Y}}_{\text{B}} + \underbrace{\int_{\mathcal{X}} uv \, \mathrm{d}\mathcal{X}}_{\text{C}} = -\underbrace{\int_{\mathcal{X}} p \, \mathrm{d}\mathcal{X}}_{\text{D}} + \underbrace{Re^{-1} \left(\int_{\mathcal{Y}} \partial_x v \, \mathrm{d}\mathcal{Y} + \int_{\mathcal{X}} \partial_y v \, \mathrm{d}\mathcal{X} \right)}_{\text{E}}. \tag{7}$$

Using the same discretization steps, we may integrate and divide by $\Delta x \Delta y$ as to obtain

$$\begin{aligned}
&\partial_t v_{i,j} \\
&+ \frac{1}{4\Delta y} \left[\left(v_{i,j+1} + v_{i,j}\right)^2 - \left(v_{i,j} + v_{i,j-1}\right)^2 \right] \\
&+ \frac{1}{4\Delta x} \left[\left(u_{i+1,j} + u_{i,j}\right)\left(v_{i,j} + v_{i+1,j}\right) - \left(u_{i-1,j} + u_{i,j}\right)\left(v_{i-1,j} + v_{i,j}\right) \right] \\
&= -\frac{p_{i,j+1} - p_{i,j}}{\Delta y} + Re^{-1} \left(\frac{v_{i+1,j} - 2v_{i,j} + v_{i-1,j}}{\Delta x^2} + \frac{v_{i,j+1} - 2v_{i,j} + v_{i,j-1}}{\Delta y^2} \right)
\end{aligned} \tag{8}$$

1.3 Continuity equation

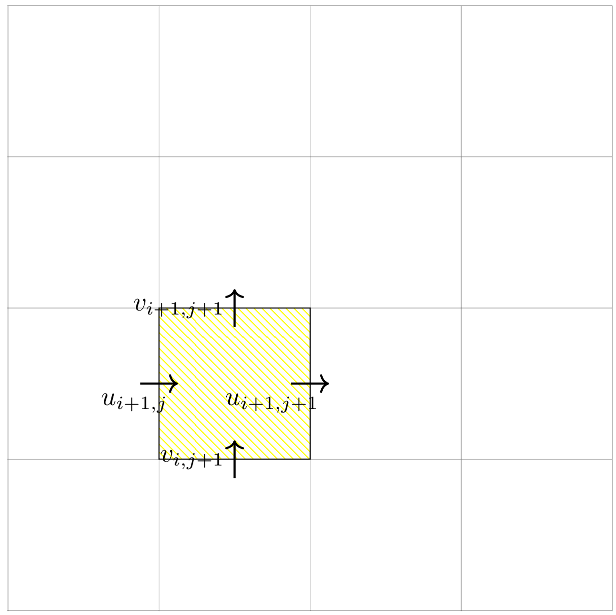

Figure 4: Control volume for the continuity equation.

Starting from equation (4), and using the control volume in figure 4, we have

$$(u_{i+1,j+1} - u_{i+1,j})\,\Delta y + (v_{i+1,j+1} - v_{i,j+1})\,\Delta x = 0. \tag{9}$$

Dividing through by Δx and Δy, we obtain

$$\boxed{\frac{1}{\Delta x}\left(u_{i+1,j+1} - u_{i+1,j}\right) + \frac{1}{\Delta y}\left(v_{i+1,j+1} - v_{i,j+1}\right) = 0.}$$